

2-Mavzu: PyQt5 paketi va QtDesigner dasturi yordamida GUI dasturlarini yaratish.

8-mashg‘ulot. Matn muharriri dasturini yozish.

O‘quv savollari:

1. PyQt5 da matn muharriri dizaynidagi elementlarning funksiyalari uchun dastur yozish.
2. Dasturni testlash.

1. PyQt5 da matn muharriri dizaynidagi elementlarning funksiyalari uchun dastur yozish.

```
import sys
from PyQt5 import uic
from PyQt5.QtWidgets import (
    QApplication, QMainWindow, QFileDialog, QMessageBox,
    QLabel, QTextEdit, QPlainTextEdit, QAction
)

class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()

        # 1) .ui dizaynini yuklash (Qt Designer'da chizilgan interfeys)
        uic.loadUi("TextEditor.ui", self)

        # 2) Muhim vidjetlar: matn maydoni va status label (Ln/Col)
        # Sizning .ui faylingizda QTextEdit nomi: "textEdit"
        self.text_edit = (self.findChild(QTextEdit, "textEdit")
                           or self.findChild(QPlainTextEdit, "textEdit")
                           or self.findChild(QTextEdit, "plainTextEdit")
                           or self.findChild(QPlainTextEdit, "plainTextEdit"))
        self.lbl_lncol: QLabel = self.findChild(QLabel, "lblLnCol")

        # 3) Holat: joriy fayl yo‘li va "o'zgargan/o'zgarmagan" flag
        self.current_path = None
        self.is_modified = False

        # 4) QAction'larni topish (sizdagi .ui da ba'zilarida trailing space bor)
        def A(name: str):
            return self.findChild(QAction, name)

        self.act_new = A("actionNew ") or A("actionNew")
        self.act_open = A("actionOpen")
        self.act_save = A("actionSave ") or A("actionSave")
        self.act_save_as = A("actionSaveAs")
        self.act_exit = A("actionExit")

        self.act_undo = A("actionUndo")
        self.act_redo = A("actionRedo")
        self.act_cut = A("actionCut")
        self.act_copy = A("actionCopy")
```

```

self.act_paste      = A("actionPaste")
self.act_selectall  = A("actionSelectAll")

self.act_about      = A("actionAbout")

# 5) QAction → funksiyalarga ulash (File)
if self.act_new:      self.act_new.triggered.connect(self.new_file)
if self.act_open:     self.act_open.triggered.connect(self.open_file)
if self.act_save:     self.act_save.triggered.connect(self.save_file)
if self.act_save_as:  self.act_save_as.triggered.connect(self.save_file_as)
if self.act_exit:     self.act_exit.triggered.connect(self.close)

# 6) QAction → funksiyalarga ulash (Edit)
if self.act_undo:     self.act_undo.triggered.connect(lambda:
self.text_edit and self.text_edit.undo())
if self.act_redo:     self.act_redo.triggered.connect(lambda:
self.text_edit and self.text_edit.redo())
if self.act_cut:      self.act_cut.triggered.connect(lambda:
self.text_edit and self.text_edit.cut())
if self.act_copy:     self.act_copy.triggered.connect(lambda:
self.text_edit and self.text_edit.copy())
if self.act_paste:    self.act_paste.triggered.connect(lambda:
self.text_edit and self.text_edit.paste())
if self.act_selectall: self.act_selectall.triggered.connect(lambda:
self.text_edit and self.text_edit.selectAll())

# 7) QAction → funksiyalarga ulash (Help)
if self.act_about:    self.act_about.triggered.connect(self.show_about)

# 8) Matn o'zgarsa va kursor joyi o'zgarsa → holatlarni yangilash
if self.text_edit:
    self.text_edit.textChanged.connect(self.on_text_changed)
    self.text_edit.cursorPositionChanged.connect(self.update_ln_col)

# 9) Boshlang'ich ko'rinish
self.update_window_title()
self.update_ln_col()

# ----- Yordamchi: oyna sarlavhasi -----
def update_window_title(self):
    name = self.current_path if self.current_path else "Untitled"
    if self.is_modified:
        name += " *"
    self.setWindowTitle(f"Sodda matn muharriri – {name}")

# ----- Yordamchi: Ln/Col indikator -----
def update_ln_col(self):
    if not self.text_edit or not self.lbl_lncol:
        return
    c = self.text_edit.textCursor()
    line = c.blockNumber() + 1

```

```

col = c.positionInBlock() + 1
self.lbl_lncol.setText(f"Ln {line}, Col {col}")

# ----- Yordamchi: matn olish/joylash -----
def get_text(self):
    return self.text_edit.toPlainText() if self.text_edit else ""

def set_text(self, text: str):
    if self.text_edit:
        self.text_edit.blockSignals(True)
        self.text_edit.setPlainText(text)
        self.text_edit.blockSignals(False)
        self.is_modified = False
        self.update_window_title()
        self.update_ln_col()

# ----- Yordamchi: saqlashdan oldin so'rash -----
def confirm_discard_changes(self) -> bool:
    if self.is_modified:
        reply = QMessageBox.question(
            self, "Saqlansinmi?",
            "Saqlanmagan o'zgarishlar mavjud. Saqlashni xohlaysizmi?",
            QMessageBox.Yes | QMessageBox.No | QMessageBox.Cancel
        )
        if reply == QMessageBox.Yes:
            return self.save_file()
        elif reply == QMessageBox.Cancel:
            return False
    return True

# ----- Matn o'zgartirilganda -----
def on_text_changed(self):
    self.is_modified = True
    self.update_window_title()

# ===== FILE MENYUSI =====
def new_file(self):
    # Yangi hujjat: saqlanmagan o'zgarishlar bo'lsa - so'raysiz
    if not self.confirm_discard_changes():
        return
    self.current_path = None
    self.set_text("")

def open_file(self):
    # Fayl ochish: dialog, UTF-8 ustun, bo'lmasa system default
    if not self.confirm_discard_changes():
        return
    path, _ = QFileDialog.getOpenFileName(
        self, "Open", "", "Text Files (*.txt);;All Files (*.*)"
    )
    if not path:

```

```

        return
    try:
        with open(path, "r", encoding="utf-8") as f:
            content = f.read()
    except UnicodeDecodeError:
        with open(path, "r", encoding=sys.getdefaultencoding(),
errors="replace") as f:
            content = f.read()
    self.current_path = path
    self.set_text(content)

def save_file(self) -> bool:
    # Saqlash: yoʻl bor boʻlsa shu faylga, boʻlmasa "Save As"
    if self.current_path is None:
        return self.save_file_as()
    try:
        with open(self.current_path, "w", encoding="utf-8") as f:
            f.write(self.get_text())
        self.is_modified = False
        self.update_window_title()
        self.statusBar().showMessage("Saved", 2000)
        return True
    except Exception as e:
        QMessageBox.critical(self, "Xatolik", f"Saqlashda xatolik:\n{e}")
        return False

def save_file_as(self) -> bool:
    # "Save As": yangi nom/yoʻl tanlash
    path, _ = QFileDialog.getSaveFileName(
        self, "Save As", "", "Text Files (*.txt);;All Files (*.*)"
    )
    if not path:
        return False
    self.current_path = path
    return self.save_file()

# ===== HELP MENYUSI =====
def show_about(self):
    QMessageBox.information(
        self, "About",
        "Sodda matn muharriri (PyQt5)\n"
        "Qt Designer'da chizilgan .ui asosida ishlaydi."
    )

def main():
    app = QApplication(sys.argv)
    win = MainWindow()
    win.show()
    sys.exit(app.exec_())

if __name__ == "__main__":

```

```
main()
```

Kodning punktma-punkt izohi (qisqa):

- `uic.loadUi(...)` – Qt Designer’da chizilgan UI’ni to‘g‘ridan-to‘g‘ri yuklaydi.
- `findChild(QTextEdit, "textEdit")` – matn maydonini UI’dan topadi (sizda `textEdit`).
- **QActionlarni topish** – `.ui`dagi nomlar bo‘yicha (trailing space borlarini ham) topib, `.triggered.connect(...)` bilan funksiyalarga ulaymiz.
- **File amallari** – New/Open/Save/Save As/Exit: dialoglar orqali fayl bilan ishlash, sarlavha va statusni yangilash.
- **Edit amallari** – Undo/Redo/Cut/Copy/Paste/Select All: `QTextEditing` tayyor metodlari.
- **StatusBar** – `lblLnCol` orqali **Ln/Col** doimiy yangilanadi. Saqlaganda “Saved” yozuvi chiqadi.
- **Modified flag** – `textChanged` signali bilan * belgisi sarlavhada ko‘rinadi.

2) Dasturni testlash

A. Muhit tayyorlash

1. `venv`ni yoqing:
`D:\...\Test\venv\Scripts\activate`
2. O‘rnatish:
`pip install PyQt5`
3. Dastur papkasida `.ui` va `main.py` yonma-yon tursin. `UI_PATH` dagi nom mos bo‘lsin.

B. Ishga tushirish

- `python main.py`

C. Interfeys testi (Preview emas, real ish)

1. **Ochiq oynani tekshiring:** sarlavha Sodda matn muharriri – Untitled.
2. **Menu/Toolbar** ko‘rinadimi? **StatusBar**da Ln 1, Col 1 bormi?

D. File menyusi testlari

1. **New:** matn yozing → New. “Saqlanadimi?” savoli chiqishi kerak.
2. **Open...:** `.txt` tanlang → kontent oynaga yuklansin; sarlavhada fayl nomi ko‘rinsin; Ln/Col ishlasin.
3. **Save:** fayl yo‘li bo‘lsa saqlansin, pastda “Saved” xabari ko‘rinsin; yulduzcha (*) yo‘qolsin.
4. **Save As...:** yangi nom bering → saqlansin, sarlavha yangilansin.
5. **Exit:** o‘zgartirish bo‘lsa, “Saqlanadimi?” degan savol chiqishi kerak.

E. Edit menyusi testlari

1. **Undo/Redo:** matn yozib, bekor qilish/ortga qaytarishni sinang.
2. **Cut/Copy/Paste:** tanlab kesish/nusxa/joylash ishlasin.
3. **Select All:** butun matn ajratilsin.

F. StatusBar testi

1. Kursorni turli joyga olib boring → **Ln/Col** to'g'ri yangilansin.
2. Saqlagandan keyin **“Saved”** yozuvi 2 soniya ko'rinsin.

G. Burchak holatlari

1. **Bo'sh saqlash:** `Untitled` holatda darrov `Save` → `Save As` dialogi ochilishi kerak.
2. **Open** → **Cancel:** hech narsa o'zgarmasin.
3. **Kodlash:** UTF-8 bo'lmagan fayl ochsangiz ham matn chiqishi kerak (noan'anaviy belgilar o'rnida  bo'lishi mumkin — bu kutilgan).